MODULE 13

Investigation with Splunk and ELK

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What You Will Learn

Logs are the computer's diary that **does not lie**. This module makes you fluent in reading them.

- **What logs are** - and why courts trust them more than human memory
- **Splunk Free** - install it, ingest logs, search like Google
- **SPL** - five searches that catch failed logins, scanners and thieves
- **ELK stack** - Elasticsearch, Logstash, Kibana and cloud dashboards
- **Timelines and alerts** - from raw logs to court-ready story
- **Legal handling** - hashes, timezones, BSA s.63 certificates

Logs: The Diary That Does Not Lie

Every computer writes a diary: **who came, when, what they did**. Humans forget; logs don't:

- A witness says 'I did nothing' - the logs show what that user **actually did** at that hour
- Even a small server writes **lakhs of lines daily** - the criminal cannot erase them all
- For an investigator, logs are **digital witnesses** - silent, exact, and always present

Three Reasons Logs Are Gold

Why do investigators and courts trust logs above almost everything else?

- **Automatic:** no human writes them - no forgetting, no lying, no bias
- **Timestamped:** every event carries date-time - the whole incident becomes a timeline
- **Abundant:** lakhs of lines per day - somewhere, some log **always survives**

Five Log Types: Know Where to Look



- Account hacked? Start with the **auth log** - failed logins tell the story
- Website attacked? The **web log's 404s** reveal the reconnaissance

Splunk: Google for Logs

Millions of log lines, one search box. **Splunk finds the needle in the haystack**:

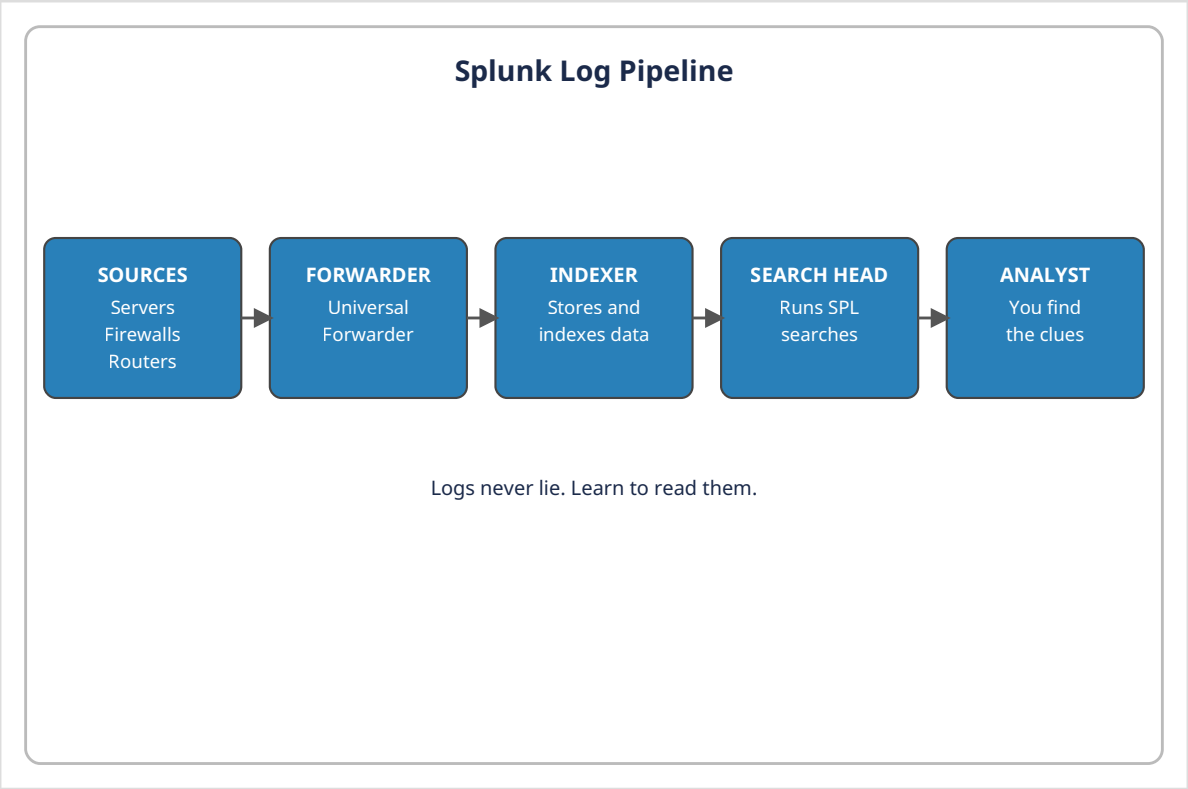
- Free version handles **500 MB/day** - plenty for training and small cases
- Two parts to know: the **forwarder** (collects logs from machines) and the **index** (organised store)
- Think: forwarder = reports from every thana; index = the record room at HQ

Forwarder and Index: The Two Pillars

Understand these two and Splunk stops being mysterious:

- **Universal forwarder:** small software on each machine - ships its logs to central Splunk
- **Index:** the named store where logs live - you always search **inside an index**
- In every search you first say **where**: `index=weblogs`

Splunk Log Pipeline



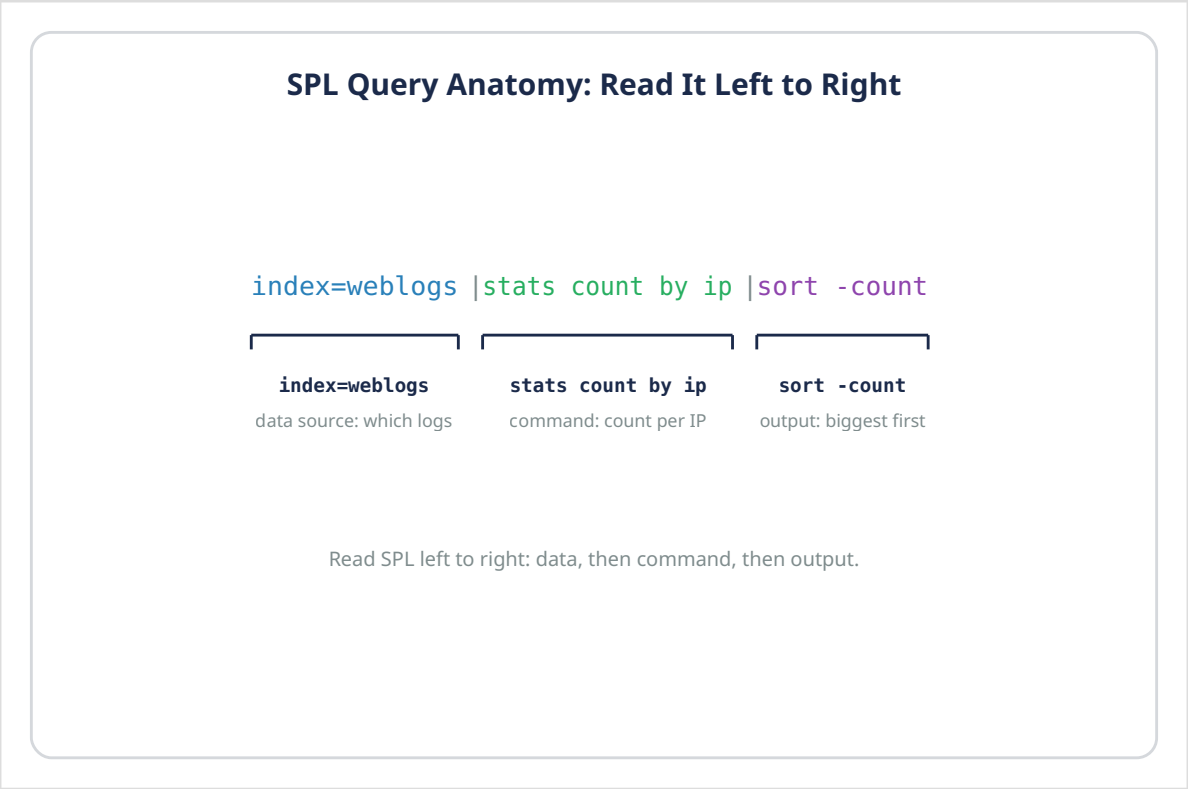
- Machines → forwarder → index → **your SPL search** → timeline
- The pipeline is the same in Splunk and ELK - only the names change

SPL: Read It Left to Right

SPL (Search Processing Language) is **easier than it looks**. Three rules:

- Read **left to right**: first WHERE to search, then WHAT to do
- The **pipe |** means 'send the result ahead' to the next command
- Example: `index=weblogs | stats count by clientip`

SPL Query Anatomy



- Three labelled parts: **data source | filter/command | output**
- Master this one pattern and you can read any SPL search

Five SPL Searches That Catch Criminals

Copy these into your notebook - they solve **most log investigations**:

- Failed logins: `index=weblogs status=401 OR status=403 | stats count by clientip`
- Top talkers: `index=weblogs | top limit=10 clientip`
- Recon scan: `index=weblogs status=404 | stats count by uri_path | sort - count`

Time Windows and Clean Tables

Two more tools that turn searches into **evidence**:

- Time range: `index=weblogs earliest="10/14/2026:02:10:00" latest="10/14/2026:02:20:00"`
- Clean table: `index=weblogs clientip=203.0.113.77 | table _time, uri_path, status, bytes`
- **table** shows every event; **stats** counts them - use both

Six-Step Log Investigation



- Step 1 decides everything: **no question = hours wasted** in the ocean of logs
- Secure logs **first**, analyse later - servers auto-delete old logs

ELK: The Three-Part Stack

Splunk's free alternative - **three tools, one pipeline**, cloud-friendly:

- **Elasticsearch:** stores massive data, searches at lightning speed
- **Logstash:** collects logs from everywhere, makes them uniform - one language
- **Kibana:** the dashboard - graphs, charts, the picture the SP understands

Kibana: Charts That Convince

A bar chart of 'top IPs' tells the story **faster than 1,000 log lines**:

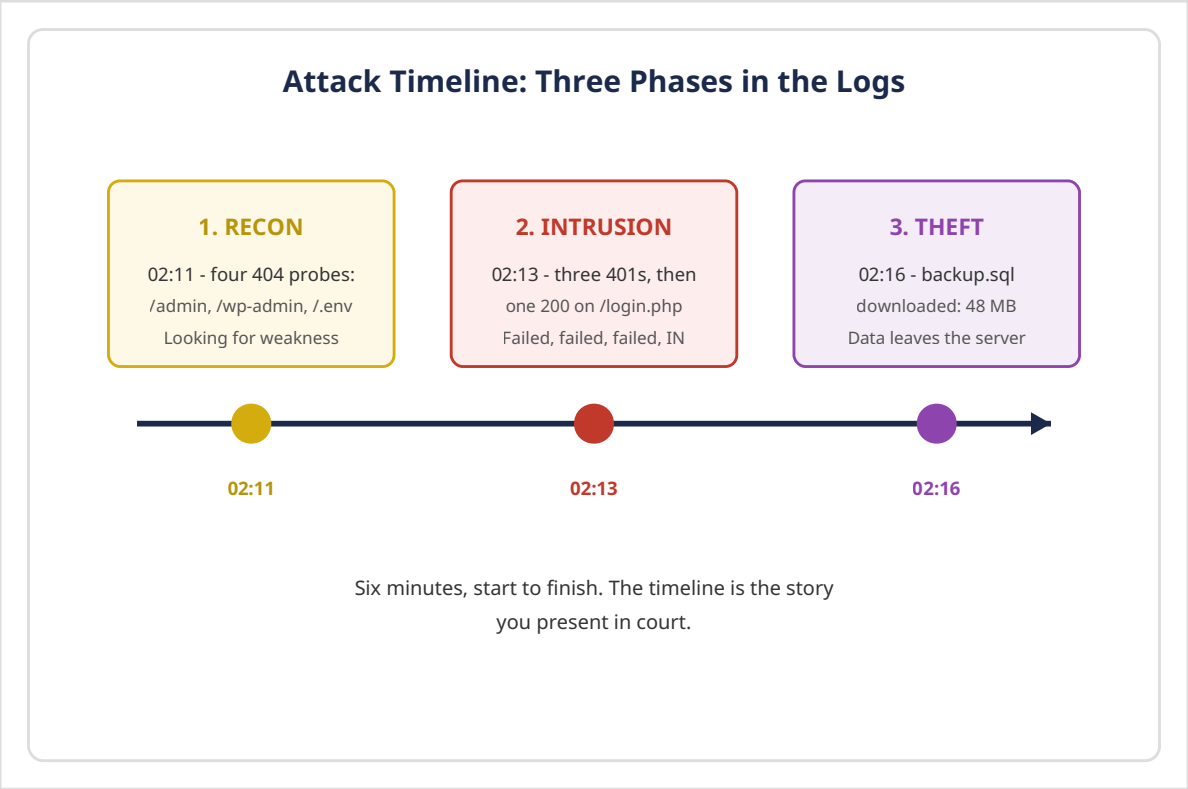
- Upload logs → create data view → bar chart on clientip → the attacker's bar towers over all
- Time filter is your most-used tool: narrow to the incident window
- Save the dashboard - it becomes **exhibit material** for the case file

Alerts: Guard Duty, Automated

Why wait for a complaint? **Teach the system to shout** when something smells wrong:

- '10 failed logins from one IP in 5 minutes - **alert me immediately**'
- Midnight logins, giant data transfers, dead accounts waking up - all alert-worthy
- Many big attacks were caught **early** by alerts; investigation starts before damage

Attack Timeline: Three Phases



- **Recon → intrusion → theft:** almost every attack follows this arc
- Six minutes start-to-finish in the sample - speed is why you secure logs **first**

Log Evidence: Four Court Rules



- The defence will ask: 'prove these logs are genuine' - **have answers ready**
- Hash + timezone + certificate + written steps = **unshakeable**

BSA s.63: The Certificate

Digital evidence needs a **certificate** to enter the courtroom:

- Officer certifies: records created in **normal working**, not tampered with
- Earlier: IT Act s.65B - now: **BSA s.63** (Bharatiya Sakshya Adhiniyam, 2023)
- Prepare it at the **start** of investigation - last-minute paperwork looks shaky

Four Mistakes Even Seniors Make

Memorise these - they separate **professionals from amateurs**:

- **Drowning in data:** searching without a question and time range - hours wasted
- **Forgetting timezone:** UTC vs IST = timeline wrong by **5.5 hours**
- **Alert as evidence:** the alert hints; the **log lines** are the evidence
- **Late log seizure:** log rotation deletes old logs - secure first, analyse later

Real Case: AIIMS Ransomware, Nov 2022

India's top hospital hit by ransomware - **logs led the investigation**:

- Servers encrypted, patient care on paper for weeks - national-level crisis
- Agencies dug through server logs: unusual logins, changed files, data exfiltration paths
- Lesson: an institution's **logs are its black box** - protect them before disaster
- Source The Indian Express

Real Case: WannaCry, 2017

Ransomware that crippled lakhs of computers worldwide - **logs mapped its spread**:

- Hospitals, power departments, small offices across India affected
- Network logs revealed the infection route: **which machine first, then how it jumped**
- That map shaped global defences. Lesson: logs don't just solve crimes - **they prevent the next one**
- Source The Hindu

Your Turn: Labs in This Module

Four labs - from **zero to timeline**, all on your own machine:

- **Lab 1:** Install Splunk Free, ingest the 19-line sample log - count must read 19
- **Lab 2:** Write and run 5 SPL searches + the cheat sheet in your notebook
- **Lab 3:** Build a Kibana bar chart of top IPs on Elastic Cloud's free trial
- **Lab 4:** Build the 3-phase incident timeline and export it as CSV for the case file

Key Takeaways

If you remember nothing else from Module 13:

- Logs are the **diary that does not lie** - automatic, timestamped, abundant
- Splunk = **Google for logs**: forwarder collects, index stores, SPL searches
- Read SPL **left to right**: where, pipe, what - five searches solve most cases
- Investigate in six steps: **question, log, time, pattern, timeline, preserve**
- Court needs: **hash, timezone, BSA s.63 certificate, written steps**
- Secure logs **first** - analysis later; rotation deletes while you wait

Review Questions

- 1. Give three reasons logs are more reliable than human witnesses, in your own words.
- 2. Explain: `index=weblogs status=404 | stats count by uri_path` - what does each part do?
- 3. A server was attacked at 2 AM. List your six investigation steps in order.
- 4. The log says 02:11 but your watch says 07:41. What went wrong, and how do you fix it?
- 5. Why is an alert not evidence? What do you file in the case diary instead?

NEXT: MODULE 14

ATM and Card Crime Investigation

Skimmers, cloning, impossible travel - follow the information